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SPORTS PERFORMANCE

Kinetics and kinematics of initial sprint acceleration performance in Division I collegiate American football players

Cameron M. Josse[a], Jean-Benoit Morin[a], Kenneth P. Clark[b], Joel T. Greenshields[c] and Aaron D. Wellman[d]

[a]Inter-University Laboratory of Human Movement Biology, University Jean Monnet, Saint-Etienne, France; [b]Department of Kinesiology, West Chester University, West Chester, Pennsylvania, USA; [c]Department of Sports Science, New York Mets, Queens, New York, USA; [d]Department of Player Performance, New York Football Giants, East Rutherford, New Jersey, USA

ABSTRACT

This study collected the first direct measurements of the kinetic and kinematic variables associated with initial acceleration performance in American football players. 123 male NCAA Division I American football players performed one-to-two maximal sprints of 20–30-meters. Kinetics were derived from a linear motorised encoder and individual linear sprint force-velocity-power (FVP) profiles were calculated utilising a validated field method. Kinematics were derived from video images recorded in slow-motion. The first three steps, including the initial push-off, were used for analysis at the instants of toe-off and touchdown. Large to nearly perfect, significant relationships were obtained between maximal power and 10-meter time. The spatiotemporal variables with the strongest relationships to maximal power were contact time, takeoff and step velocity, and normalised flight length. At toe-off, the swing leg hip angle, stance leg thigh angle, and CM-toe distance had large relationships with maximal power. At touchdown, the shank angles of both the stance and swing leg and the stance leg knee angle had large relationships. Maximal power associated with force output in the horizontal direction may considered a key performance indicator for initial acceleration in American football and certain technical underpinnings are associated with higher levels of maximal power during initial sprint acceleration.

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Force; velocity; power; running

Introduction

American football is a high-intensity, intermittent sport where linear speed capabilities are some of the most sought-after physical performance parameters. When scouting or recruiting prospective players, acceleration is a highly sought-after skill. Players that performed better in the National Football League (NFL) Scouting Combine 40-yard (36.6-m) dash test were typically drafted higher than their slower counterparts across different position groups (Sierer et al., 2008). In football, each play consists of repeated short, high intensity actions that last typically between 5–6 s per play but may range anywhere from 2–13 s. On average, it has been shown that there are 6–8 plays per series and 11–14 series per game (Hoffman, 2008; Rhea et al., 2006). Therefore, players must possess the ability to accelerate at a high level and do so over the course of several plays throughout a game. The in-game average sprinting (> 23.0 km/h) (Jessiman et al., 2019; Konarski et al., 2023; Schwesig et al., 2019; Wellman et al., 2016) distances are associated with player positions and can range from less than 10-m in as little as one sprint effort to well over 300-m in over 10 sprint efforts (Wellman et al., 2016). Additionally, maximal acceleration efforts (> 3.5 m/s^2) may range from less than 5 efforts to over 20 efforts in a game.

The running biomechanics underlying sprint performance typically fall into one of two macroscopic areas: the kinetics (i.e., mechanical determinants of body movement by the fundamental laws of motion applied to the centre of mass) and the kinematics (i.e., technical execution, overall segment angles and spatiotemporal step variables). Thus, formulating mechanical and technical profiles of sprint performance may be useful to describe speed capabilities in more detail. Analysing the force-velocity-power (FVP) relationship during an all-out maximal sprinting effort is one method to profile mechanical output from a macroscopic (centre of mass) standpoint. In addition to reference force-plate studies (Rabita et al., 2015), simple methods for calculating this relationship have been validated against laboratory measurements (J. B. Morin et al., 2019; Samozino et al., 2016) and incorporated primarily into the analysis of sprint performance for track athletes, soccer players, and rugby players (Cross et al., 2015; Mendiguchia et al., 2016; Slawinski et al., 2017). For example, these calculations were used to analyse the sprint acceleration force-velocity relationship in 553 total participants from

CONTACT Cameron M. Josse cameronjosse@gmail.com 25475 Dundee Rd, Royal Oak, Michigan 48067, USA
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14 different sport disciplines and all levels of practice (Jiménez-Reyes et al., 2018). However, to our knowledge, information available for FVP profiles associated with elite American football players is absent from the peer-reviewed literature. An analysis of sprint acceleration mechanical profiling for the NFL draft was performed, but the major limitation was an indirect measurement protocol whereby the authors utilised split time data from freely available online resources with no direct measurement of raw body mass, position- or velocity-time input data, and other forms of input data required for maximum validity (Delaney et al., 2018). Additionally, the available split time data (NFL combine testing) have not been associated with consistent fully automatic timing, which may have further misrepresented the sprint performance measures and outputs. As a result, FVP profiling in American football requires more descriptive and recent data based on valid inputs for the calculation methods that are now widely used in high-performance sport science.

Combining kinetic and kinematic analyses is useful in understanding the underlying joint segmental and spatiotemporal aspects of sprint performance. Information associated with the kinematics surrounding the stance and swing phases of each leg and their interdependence during the instant of toe-off and touchdown may provide practitioners with key performance indicators based on external movement expression. For example, kinematic associations with early acceleration performance were analysed in a variety of field sport athletes and it was found that subjects with faster horizontal velocity in early acceleration had significantly lower contact times and an increased stride frequency, with no difference found in stride length (Murphy et al., 2003). Another analysis had similar findings, with step frequency and step time showing the greatest level of association with 10-m sprint performance, although this association was described as 'small' (Standing & Maulder, 2017). A comparison was made between rugby forwards, backs and sprinters, finding that sprinters achieved substantially greater levels of acceleration performance, as determined by normalised average horizontal external power (NAHEP) (Wild et al., 2018). The primary technical variable that showed the strongest association with NAHEP was toe-off distance, defined as the distance between the toe and centre of mass (CM) at the instant of toe-off. Additionally, the relationships between various kinematics and mean ratio of force (RFmean), described as the proportion of the stance-averaged resultant ground reaction force (GRF) vector directed horizontally, were investigated (King et al., 2022). Specifically, the authors related RFmean to kinematic variables retrieved from the first four steps of a sprint from blocks with collegiate sprinters, finding significant relationships for negative touchdown distance (when the foot was posterior to the centre of mass at touchdown), positive shank angles (where the knee was anterior to the foot), and higher step frequencies.

There is an interest in studying kinetic variables in addition to split times, but also the relationships between them at various segments (e.g., 0–10-m) to determine which kinetic variables relate to sprint performance over targeted distances. For example, the same split times can result from different underlying mechanics and FVP profiles and the extent to which the profile orientation impacts sprint performance is based on individual values of Pmax and the targeted sprint distance (e.g., 0–10- vs. 0–20-m) (Samozino et al., 2022). As a further layer, there is interest in seeing how these relationships differ between player position groupings because the differences may provide more guidance in determining key performance indicators of initial acceleration performance as they relate to playing position. To our knowledge, the description of collective kinetics and kinematics associated with initial acceleration in high-level American football have not been published. Additionally, it is important to investigate how kinetics and kinematics may differ by player position in their influence over initial acceleration performance due to different combinations of body mass and height. The aim of the present pilot study was thus (a) to bring the first direct measurements of macroscopic mechanical and technical variables associated with the initial steps of acceleration in NCAA Division I college football players by positional groups and (b) to examine how each variable of interest relates to sprint performances over 0–10-m.

Methods

Experimental approach

Subjects

One hundred and twenty-eight male NCAA Division I American football players (mean ± SD; age: 21 ± 2 years; height: 1.86 ± 0.1 m; body mass: 101.5 ± 20.0 kg) performed sprint performance testing as part of their scheduled off-season training routine. The players were categorised based on position: offensive line (OL), defensive line (DL), outside linebacker (OLB), inside linebacker (ILB), tight end (TE), quarterback (QB), running back (RB), wide receiver (WR), and defensive back (DB). All participants were free from injury and trained 4–5 days a week at the time of data collection. The participants ranged in experience from players that were recently enrolled in the university (within the prior 2–4 months) to players

who had been attending the university for over 4 years. Thus, training history and sprint training experience was varied. However, at the time of testing, all participants had been involved in sprint-based training for a minimum of four weeks. All data were collected on two separate testing dates, approximately one year apart, both during the winter-spring off-season training period. Each testing period featured a separate cohort of players, with 73 total players in the first cohort and 55 players in the second cohort. While both cohorts were qualified according to the sprint-based training experience described previously, the specific training interventions were not identical from one year to the next which may have influenced the results. This study received ethical approval through the Auburn University Institutional Review Board under the category of exempt research. According to 45 CFR 46.104, this study qualifies for exemption due to involving performance-based assessments conducted in controlled athletic performance environments, where players were evaluated through standardised routine testing. While formal written consent was not required, players were informed of the purpose of the testing, the procedures, and the voluntary nature of participation. Agreement to participate was obtained by all players included in this study and all data collected were de-identified to ensure confidentiality.

Data collection

Sprint performance testing

Participants performed a standardised dynamic warm-up that lasted approximately twenty minutes and consisted of exercises devoted to heart rate elevation, ground-based and transitional dynamic mobility, rhythmical activities (e.g., skips, hops, and bounds), as well as various accelerations from dynamic and static starting positions to prepare for the sprint testing. To properly calculate sprint FVP profiles, players were required to sprint a distance over which they would reach at least 94% of their maximal speed, and it has been found that even a collection of the fastest sprinters in the world were shown to achieve more than 94% of relative maximal speed between 30–40-m (Krzysztof & Mero, 2013). One-to-two maximal sprints of 20–30-m were performed, with the prescribed distance based on player position. The OL and DL players were more likely to reach a higher percentage of their maximal speed over shorter distances compared to the other positions, since they had significantly larger body mass (Table 1) and represented a population that has historically been much slower than other positions (Brechue et al., 2010; Clark et al., 2019; Fry & Kraemer, 1991; Sierer et al., 2008). Therefore, these players performed a maximal 20-m sprint while all other positions sprinted for 30-m. Some participants declined to perform a second sprint due to feelings of tightness in the legs and did not want to risk potential strain or other injury. The percentage of subjects who only completed one trial was 20.1% of the total sample size. Of the players who performed two sprint trials, intraclass correlation coefficient (ICC) results indicated excellent reliability between trials. The single measure ICC was 0.96 (95% CI: 0.92–0.98) and the average measure ICC was 0.98 (95% CI: 0.96–0.99), both indicating excellent reliability between sprint trials.

The kinetic outputs and mechanical efficiency were taken from the best time trial. If a participant ran the same time on both trials, the trial with the higher maximal power (Pmax in W/kg) in the antero-posterior direction was used in the analysis. Each sprint trial was

Table 1. Anthropometric, sprint time, FVP profile, and spatiotemporal variables.

	Skill group (n = 62)			Big-Skill group (n = 37)			Lineman group (n = 24)		
	DB, WR, RB			TE, OLB, ILB, QB			OL, DL		
Variables	Mean ± SD	Min	Max	Mean ± SD	Min	Max	Mean ± SD	Min	Max
Height (m)	1.81 ± 0.06	1.68	1.98	1.90 ± 0.06†	1.78	2.01	1.93 ± 0.05†	1.85	2.01
Body mass (kg)	86.2 ± 7.32	69.0	105	106 ± 7.76†	88.0	125	134 ± 10.8†§	110	157
10 m time (s)	1.93 ± 0.07	1.75	2.04	2.00 ± 0.07†	1.80	2.12	2.14 ± 0.08†§	1.98	2.34
F0 (N/kg)	7.80 ± 0.43	6.89	8.97	7.52 ± 0.54	6.51	8.66	7.10 ± 0.65†§	5.86	8.38
Pmax (W/kg)	18.6 ± 1.39	16.0	21.3	17.0 ± 1.49†	14.3	20.0	14.3 ± 1.63†§	11.4	17.5
RFmax (%)	49.8 ± 2.93	42.7	56.0	48.8 ± 2.63	42.9	53.5	46.9 ± 3.53†§	41.9	55.0
Sfv	−0.82 ± 0.06	−0.99	−0.72	−0.83 ± 0.06	−1.00	−0.71	−0.88 ± 0.08†	−1.08	−0.75
Contact time (s)	0.19 ± 0.02	0.15	0.24	0.20 ± 0.02	0.18	0.23	0.22 ± 0.01†§	0.20	0.25
Flight time (s)	0.05 ± 0.01	0.02	0.08	0.05 ± 0.01	0.03	0.08	0.03 ± 0.01†§	0.02	0.05
Step length (m/body length)	0.62 ± 0.05	0.52	0.77	0.61 ± 0.04	0.54	0.71	0.59 ± 0.04	0.45	0.66
Step frequency (Hz)	4.23 ± 0.32	3.58	5.49	4.09 ± 0.28	3.67	4.75	3.91 ± 0.22†	3.48	4.38
Step velocity (m/s)	4.75 ± 0.31	3.93	5.29	4.75 ± 0.25	4.35	5.15	4.43 ± 0.36†§	3.25	4.90
Takeoff velocity (m/s)	4.83 ± 0.32	4.12	5.57	4.71 ± 0.26	4.21	5.19	4.25 ± 0.50†§	3.38	5.20
Contact length (m/body length)	0.45 ± 0.04	0.36	0.53	0.44 ± 0.04	0.38	0.52	0.45 ± 0.03	0.39	0.50
Flight length (m/body length)	0.13 ± 0.03	0.06	0.22	0.12 ± 0.03	0.06	0.21	0.07 ± 0.02†§	0.03	0.11

†Significant difference ($p \leq 0.05$) compared to Skill group.
§Significant difference ($p \leq 0.05$) compared to Big-Skill group.

performed from a three-point stance to coincide with the starting position of the 40-yard (36.6-m) dash test of the NFL Scouting Combine. Data were derived from a linear motorised encoder (1080 Sprint, 1080 Motion; 333 Hz, Lidingö, Sweden) set in isotonic mode with 1 kg of resistance load to reduce any slack in the cable line and the device was set 5 m behind the starting line. The cable was attached to a belt fixed around the waist of each participant. The additional 1 kg of resistance was shown not to affect inter-system comparisons for running velocity and derived data in a previous multi-system comparison that included the same device used in this study (Fornasier-Santos et al., 2022). Raw velocity data were computed at 333 Hz and each sprint trial was recorded using the auto-distance feature, automatically starting the trial recording once movement speed exceeded 0.3 m/s, which is supported by previous research as an acceptable threshold (Apte et al., 2020). Data recording ceased after surpassing the pre-programmed sprint distance (20- or 30-m). All testing was performed in an indoor training facility with an artificial turf surface and players wore cleats specific to American football. The indoor temperature (21.7 ± 4 °C) and local barometric pressure (761 ± 1.4 mmHg) were recorded to help with calculation of the sprint FVP profiling. The indoor facility featured three large overhead doors that opened to the outside, one of which was kept open throughout one of the testing periods, causing the indoor temperature to drop. Participants were provided with a rest period of at least 10 minutes between sprint trials.

Sprint force-velocity profiling

Individual linear sprint FVP profiles were calculated from raw velocity data with a validated field method based on fundamental laws of motion (Morin et al., 2019; Samozino et al., 2016). Capabilities of relative theoretical maximal force (F0 in N/kg) and Pmax were calculated for each player. It should be noted that power is a scalar quantity and has no direction, only magnitude, and it may be more appropriate to consider Pmax as a quasi-power to better describe the projection occurring during sprinting (Vigotsky et al., 2019). For the purposes of this study, the term 'maximal power' will be used to describe the apex of the power-velocity 2nd-degree polynomial relationship. Measures of mechanical effectiveness were also calculated based on the maximal ratio of forces, defined as the maximal ratio of the stance-averaged horizontal component of the GRF to the resultant force (RFmax in %). RFmax may be interpreted as the proportion of the total force production that is directed in the forward direction of motion at the start of the sprint, where a higher value indicates more of the total force output directed forward (Morin et al., 2011). Lastly, the slope of the linear force-velocity relationship (Sfv) was computed to describe the index of each player's individual balance between force and velocity capabilities. The more negative the value, the more 'force-oriented' the profile, and vice versa (Morin & Samozino, 2016).

Sprint spatiotemporal and kinematics assessment

For all sprint trials, video images were recorded in slow-motion mode (240 Hz) with a tablet camera at a high definition (HD) resolution of 1080p. The camera was placed at the 1.5-m mark, 9 m away from and perpendicular to the running lane, and set at a height of 1.1 m to allow for a field of view of approximately 9 m. The initial push-off and the subsequent 4 steps were captured. Video images were imported into Kinovea (version 2023.1) to analyse the sprint sequences of toe-off (first frame where the foot had visibly left the ground) and touchdown (first frame where the foot was visibly in contact with the ground), like previous research (Lahti et al., 2020; Wild et al., 2018, 2022). For the initial acceleration breakdown, the first three steps, including the initial push-off, were analysed and reported as averaged kinematic values. In Kinovea, the filming zone was calibrated to a 5 m horizontal distance along the midpoint of the camera along the running lane. Players' bodies were modelled as 18 points using manual digitisation at maximum (6x) zoom capability of the following: vertex of the head, halfway between the suprasternal notch and the 7th cervical vertebra, shoulder, elbow, and wrist joint centres, head of third metacarpal, hip, knee, and ankle joint centres, and the tip of the toe. All sprints were analysed at least twice to improve reliability with the digital marker method. Intra-rater reliability of the digital markers was assessed using a two-way random effect, single measure absolute agreement intraclass correlation coefficient (ICC) where the 18 markers had an average ICC value of 0.96 ± 0.03, indicating excellent reliability between analyses. The average of the values for each marker was used for further analysis. The digitalised coordinates were exported into Excel (Microsoft 365 Version 2310), allowing for computation of spatiotemporal and kinematic step characteristics. Whole-body centre of mass (CM) location was calculated using segmental data (de Leva, 1996) like other previous research (Hamner et al., 2010; Lahti et al., 2020; Virmavirta & Isolehto, 2014; von Lieres et al., 2016; Walker et al., 2021), allowing for the following calculations: CM-toe distance (horizontal distance between the CM and the toe, with negative values representing the toe behind the CM) and CM height (vertical distance

between the ground and the CM, with higher numbers indicating the CM further above the ground) (Figure 1b). The CM angle (angle formed between the CM, toe and horizontal axis) was also quantified (Figure 1a). Joint angles of the hip, knee, and ankle were quantified (Figure 1c) as well as angles of the trunk, thigh, shank, and foot relative to the horizontal line (Figure 1a). Additionally, computations of contact time (s), flight time (s), step length (m; horizontal displacement between initial contact of one foot and the point of initial contact of the opposite foot, measured from the toe tips), and step frequency (Hz; calculated as 1/step time, where step time was determined as the sum of contact time and the subsequent aerial time), step velocity (calculated as step length multiplied by step frequency), and takeoff velocity were also included. Takeoff velocity was calculated using the following equation:

$$Takeoff\ velocity = \frac{CM_2 - CM_1}{t_2 - t_1} \quad (1)$$

where CM_1 is the horizontal position of the CM at toe-off with t_1 being the time point of toe-off and CM_2 is the

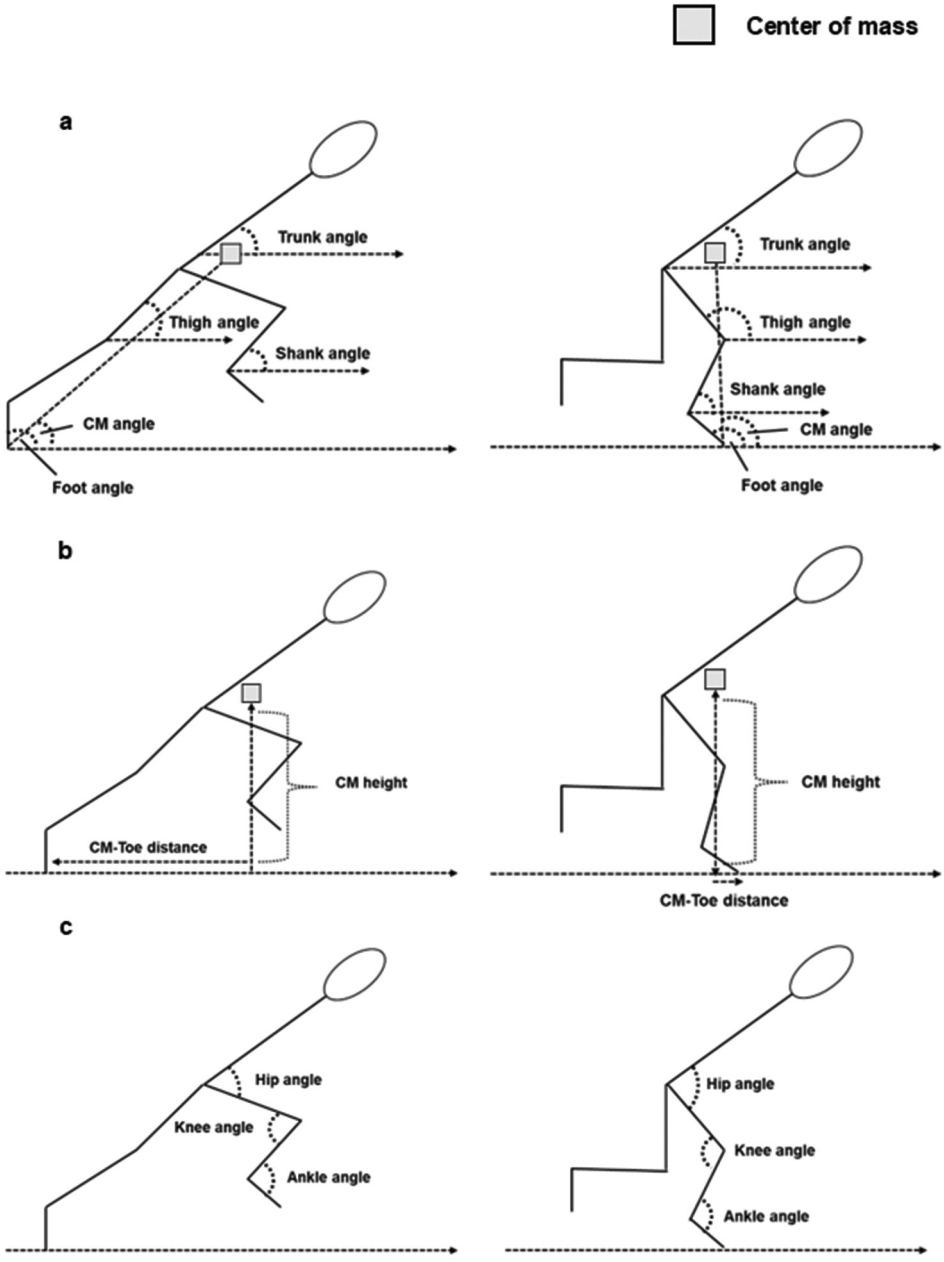

Figure 1. Kinematic step characteristics at toe-off (left panels) and touchdown (right panels): joint angles relative to the horizontal axis (a), measures of CM location relative to the horizontal and vertical axes (b), and joint angles without reference to any axis (c).

horizontal position of the CM at the subsequent touchdown with t_2 being the time point of touchdown.

All lengths and distances were normalised to the height of the athlete and reported as (m/body length). Due to a recording issue with the IOS software, 73 of the recordings were captured at only 120 Hz. These recordings were kept in the dataset, still showing good consistency of kinematic output in comparison to the subjects recorded at 240 Hz. Five players (1 OL, 2 LB, 1 RB and 1 DB) were excluded from analysis due to disruptions in the video recording, rendering kinematic analysis impossible. Therefore, the total sample size was reduced from 128 to 123 players.

Statistical analysis

Mean data for kinematic variables were obtained over the initial push-off and first three steps of each sprint trial for each participant. Descriptive statistics for all variables were presented as means, standard deviations, minimums, and maximums to indicate centrality and full range results among players. Players were placed into three broad groups based on position: Skill group (DB, WR, RB), Big-Skill group (TE, OLB, ILB, QB), and Lineman group (OL, DL). One-way ANOVAs with Bonferroni post-hoc comparisons were used to determine inter-group mean effects. Pearson product-moment correlations were used to determine the strength of the relationship between the FVP outputs and the sprint times across each group as well as the relationships between the FVP outputs and the kinematics across the whole population. Strength of relationship was based on Cohen's *d* (Cohen, 1992) thresholds, where values for *r* were interpreted as trivial (0–0.09), small (≥0.10–0.29), moderate (≥0.30–-0.49), large (≥0.50–0.69), very large (≥0.70–0.89), nearly perfect (≥0.90–0.99), and perfect (1). Confidence intervals set at 95% were calculated for each correlation. All statistical analyses were conducted using Statistical Package for the Social Sciences (SPSS for Windows, version 29.0.2.0 (20); SPSS, Inc., Chicago, IL, USA) with significance set at a value of $p < 0.05$.

Results

Anthropometrics

Significant ($p < 0.001$) main effects from ANOVA testing were obtained for body mass and height for all groups (Table 1). From post-hoc analysis, the Lineman group had significantly larger body mass than both the Big-Skill and Skill groups ($p < 0.001$). The Big-Skill group also had significantly larger body mass than the Skill group ($p < 0.001$).

Split times and FVP variables

Significant ($p < 0.001$) main effects from ANOVA testing were obtained for measures of split times and FVP variables (Table 1). To emphasise initial acceleration, only the split time from 0–10-m is included, and the Skill group had the fastest split time at this segment ($p < 0.001$). The Big-Skill and Skill groups both had significantly higher values of F0 ($p = 0.01$ and $p < 0.001$, respectively), Pmax ($p < 0.001$), and RFmax ($p = 0.01$ and $p < 0.001$, respectively) compared to the Lineman group.

Spatiotemporal and joint kinematics variables

Significant ($p < 0.001$) main effects were found for measures of spatiotemporal variables (Table 1) and main kinematic variables at toe-off (Table 2) and touchdown (Table 3). The various kinematic positions are illustrated in Figure 1. For spatiotemporal variables, the Skill and Big-Skill groups both had significantly shorter ground contact times, longer flight times, faster takeoff velocities compared to the Lineman group ($p < 0.001$). At toe-off, the Big-Skill and Skill groups had a significantly reduced CM angle (more horizontal projection angle) ($p = 0.002$ and $p < 0.001$, respectively), stance leg foot angle ($p = 0.005$ and $p < 0.001$, respectively), swing leg hip angle ($p < 0.001$), swing leg thigh angle ($p < 0.001$), and swing leg knee angle ($p < 0.001$) compared to the Lineman group. At touchdown, the Skill and Big-Skill groups had a significantly reduced stance and swing leg shank angle, as well as stance leg knee angle compared to the Lineman group ($p < 0.001$). Figure 2 illustrates the joint segmental differences between the fastest and slowest player in each group.

Relationships between FVP metrics and split times

Correlational data representing the relationships between 0–10-m split times and the FVP metrics of F0 and Pmax based on position groups can be found in Figure 3. Inter-group differences revealed that F0 had a large to very large, significant relationship with 0–10-m time for the Big-Skill ($r = -0.65$; $p < 0.001$) and Lineman ($r = -0.80$; $p < 0.001$) groups, but the relationship for the Skill group was smaller, although still significant ($r = -0.27$; $p = 0.04$). Only the Lineman group showed a significant relationship between 0–10-m time and RFmax and it was very large ($r = -0.72$; $p < 0.001$). For Pmax, each group showed a large to nearly perfect, significant relationship with 0–10-m time ($r = -0.90$ to -0.60; $p < 0.001$).

Table 2. Main kinematic variables at toe-off.

		Skill group (n = 62)			Big-Skill group (n = 37)			Lineman group (n = 24)		
		DB, WR, RB			TE, OLB, ILB, QB			OL, DL		
Variables at toe-off		Mean ± SD	Min	Max	Mean ± SD	Min	Max	Mean ± SD	Min	Max
CM-toe distance (m/body length)		−0.46 ± 0.02	−0.53	−0.40	−0.46 ± 0.02	−0.49	−0.40	−0.44 ± 0.02†	−0.47	−0.41
CM height (m/body length)		0.47 ± 0.04	0.41	0.63	0.47 ± 0.02	0.42	0.50	0.49 ± 0.01	0.47	0.51
CM angle (°)		45.5 ± 3.00	40.1	55.4	45.7 ± 1.53	41.5	49.0	48.0 ± 1.58†§	45.7	51.4
Trunk angle (°)		37.6 ± 5.88	23.6	52.8	40.4 ± 5.95	28.5	51.6	42.3 ± 5.81	33.2	54.4
Stance leg kinematics	Thigh angle (°)	56.6 ± 4.07	48.2	66.0	57.6 ± 3.77	51.9	65.6	60.4 ± 4.54†	49.4	67.6
	Shank angle (°)	29.4 ± 3.11	18.9	37.1	30.5 ± 3.08	25.6	37.7	31.9 ± 2.87†	26.7	38.5
	Foot angle (°)	75.0 ± 4.94	64.7	91.0	75.3 ± 5.38	63.9	87.0	79.7 ± 5.71†§	64.7	86.8
	Hip angle (°)	161 ± 7.20	144	178	163 ± 6.56	150	175	162 ± 7.08	149	174
	Knee angle (°)	153 ± 5.91	138	165	153 ± 5.67	140	164	152 ± 5.94	143	166
	Ankle angle (°)	134 ± 5.37	121	146	135 ± 5.21	124	144	132 ± 5.47	123	146
Swing leg kinematics	Thigh angle (°)	146 ± 4.67	135	159	145 ± 5.25	135	162	138.3 ± 3.61†§	133	144
	Shank angle (°)	57.2 ± 6.32	46.0	71.5	58.5 ± 6.24	48.0	79.9	60.9 ± 5.21	50.7	74.7
	Hip angle (°)	71.9 ± 6.16	59.4	89.1	75.4 ± 6.03	64.3	85.8	83.9 ± 7.43†§	69.1	101
	Knee angle (°)	91.5 ± 5.46	73.8	101	93.5 ± 5.41	80.7	103	102.6 ± 5.39†§	94.9	113
	Ankle angle (°)	88.8 ± 3.25	79.6	94.4	89.6 ± 3.48	82.1	95.2	92.2 ± 4.83†	85.3	101

†Significant difference ($p \leq 0.05$) compared to Skill group.
§Significant difference ($p \leq 0.05$) compared to Big-Skill group.

Table 3. Main kinematic variables at touchdown.

		Skill group (n = 62)			Big-Skill group (n = 37)			Lineman group (n = 24)		
		DB, WR, RB			TE, OLB, ILB, QB			OL, DL		
Variables at touchdown		Mean ± SD	Min	Max	Mean ± SD	Min	Max	Mean ± SD	Min	Max
Touchdown distance (m/body length)		−0.01 ± 0.03	−0.10	0.11	−0.01 ± 0.03	−0.04	0.08	−0.03 ± 0.02	−0.01	0.08
CM height (m/body length)		0.46 ± 0.04	0.41	0.60	0.46 ± 0.02	0.42	0.50	0.48 ± 0.01	0.46	0.50
CM angle (°)		91.5 ± 4.06	78.7	101	91.1 ± 3.99	84.7	101	93.4 ± 2.89	88.9	99.1
Trunk angle (°)		36.8 ± 6.41	21.5	52.1	39.6 ± 5.75	26.3	50.3	40.8 ± 6.62	26.7	52.6
Stance leg kinematics	Thigh angle (°)	128 ± 5.66	113	140	126 ± 5.88	114	141	126 ± 3.83	118	131
	Shank angle (°)	51.2 ± 4.30	42.8	59.8	52.2 ± 3.60	45.3	58.2	57.2 ± 4.32†§	47.7	65.6
	Foot angle (°)	142 ± 5.41	126	152	142 ± 5.56	130	155	146 ± 6.80	130	158
	Hip angle (°)	88.8 ± 8.09	68.9	109	93.2 ± 8.19	73.2	110	95.3 ± 8.47	82.3	113
	Knee angle (°)	103 ± 5.42	88.6	118	106 ± 5.84	93.8	122	112 ± 4.92†§	100	120
	Ankle angle (°)	89.0 ± 3.99	80.3	98	90.2 ± 4.02	80.9	97.3	91.2 ± 4.98	82.9	102
Swing leg kinematics	Thigh angle (°)	69.1 ± 7.48	55.6	88.4	70.8 ± 7.63	55.5	92.1	68.3 ± 5.19	57.1	80.6
	Shank angle (°)	12.9 ± 4.85	3.00	22.8	13.9 ± 4.73	6.33	23.3	20.0 ± 5.19†§	10.2	28.8
	Hip angle (°)	148 ± 8.97	129	164	149 ± 7.91	136	170	153 ± 8.04	126	164
	Knee angle (°)	124 ± 10.9	96.1	147	123 ± 11.6	94.3	146	132 ± 8.16§	115	146
	Ankle angle (°)	130 ± 5.30	117	141	130 ± 5.58	111	139	129 ± 5.74	121	142

†Significant difference ($p \leq 0.05$) compared to Skill group.
§Significant difference ($p \leq 0.05$) compared to Big-Skill group.

Relationships between FVP metrics, spatiotemporal variables and joint kinematics

Correlational data representing the relationships between spatiotemporal variables, joint kinematics at toe-off and touchdown, and FVP metrics are found in Table 4 for Pmax and in Table 5 for F0. See Figure 1 for illustrations of the sprint phases mentioned in this section. Only the most significant variables ($p < 0.001$) are presented. For spatiotemporal variables, contact time and flight length had large relationships with Pmax (r = −0.69 and 0.56, respectively) while takeoff velocity and step velocity had large relationships with Pmax ($r = 0.65$ and 0.58, respectively) and F0 ($r = 0.53$ and 0.59, respectively). At toe-off, the swing leg knee angle had the largest relationship with Pmax ($r = -0.67$) while the associated CM-toe distance, swing leg hip, and stance leg thigh angle had large relationships with Pmax ($r = -0.59$ to −0.54) and F0 ($r = -0.61$ to −0.52). At touchdown, the shank angle of the swing leg had a large relationship with Pmax ($r = -0.57$) and F0 ($r = -0.50$) while the shank and knee angle of the stance leg also had large relationships with Pmax ($r = -0.57$ and −0.51, respectively).

Discussion

The purpose of this study was to bring the first direct measurements of macroscopic kinetic and kinematic variables associated with the initial steps of acceleration in NCAA Division I college football players and to examine how each variable of interest relates to 0–10-m sprint performance, with a focus on position groupings. The results of the present study provide more insight into

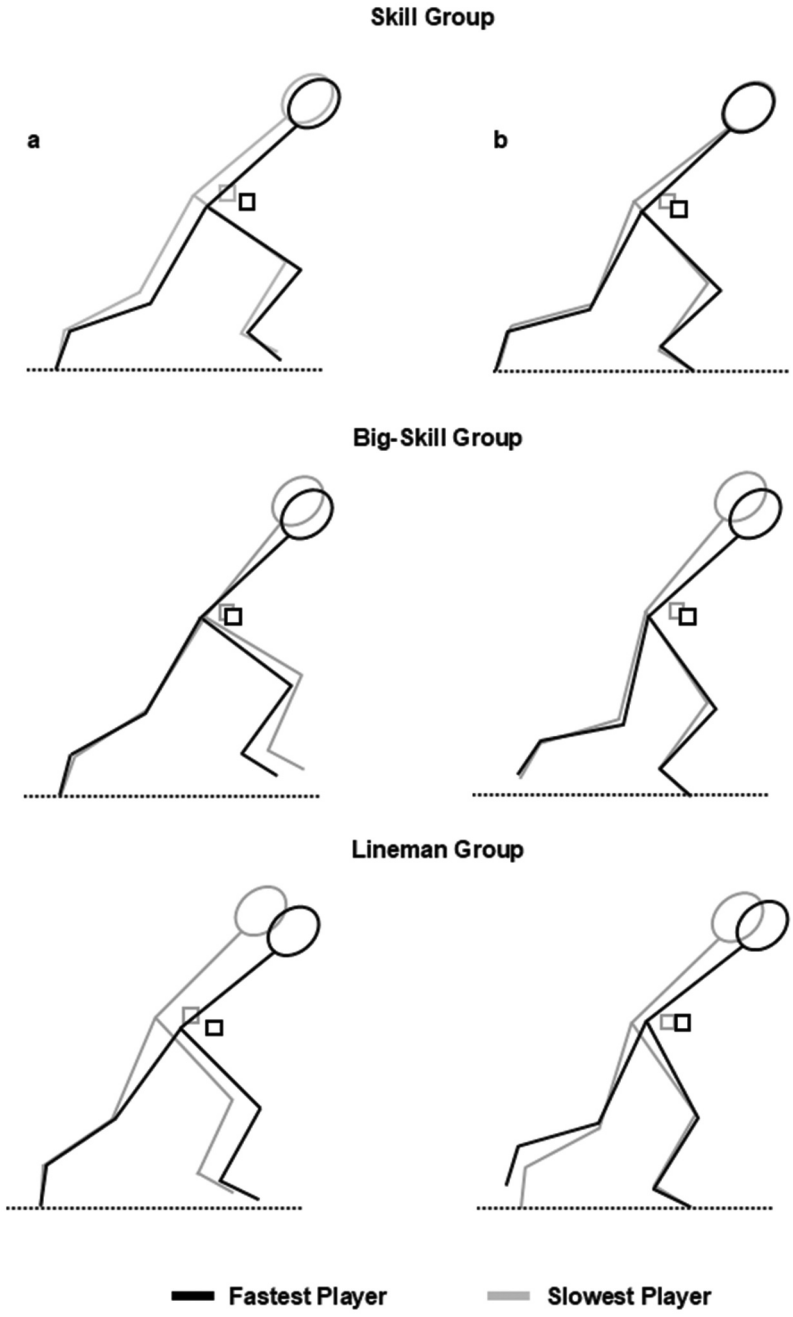

Figure 2. Joint angles and center of mass position at toe-off (a) and touchdown (b) during initial acceleration for fastest and slowest players in each position group.

the biomechanical correlates of acceleration performance in NCAA Division I college football players.

The main finding of the kinetic analysis was that Pmax had the largest relationship with the 0–10-m segment across all position groups. Thus, Pmax may serve as an appropriate key performance indicator for monitoring initial sprint acceleration. The clear relationship between Pmax and sprint acceleration performance has been shown in previous studies (Cross et al., 2017; Delaney et al., 2018; Morin et al., 2012; Rabita et al., 2015; Samozino et al., 2022), and it has been suggested that training programmes designed to improve sprint acceleration performance should focus on increasing Pmax along with the appropriate balance of F0 and/or maximum theoretical velocity (V0) (Morin & Samozino, 2016; Samozino et al., 2022). While there is no universal 'optimal' slope for a sprint FVP profile, appropriate orientation is based on the targeted sprint distance (Samozino et al., 2022). For example, for a given level of Pmax, more force-oriented profiles are appropriate for higher levels

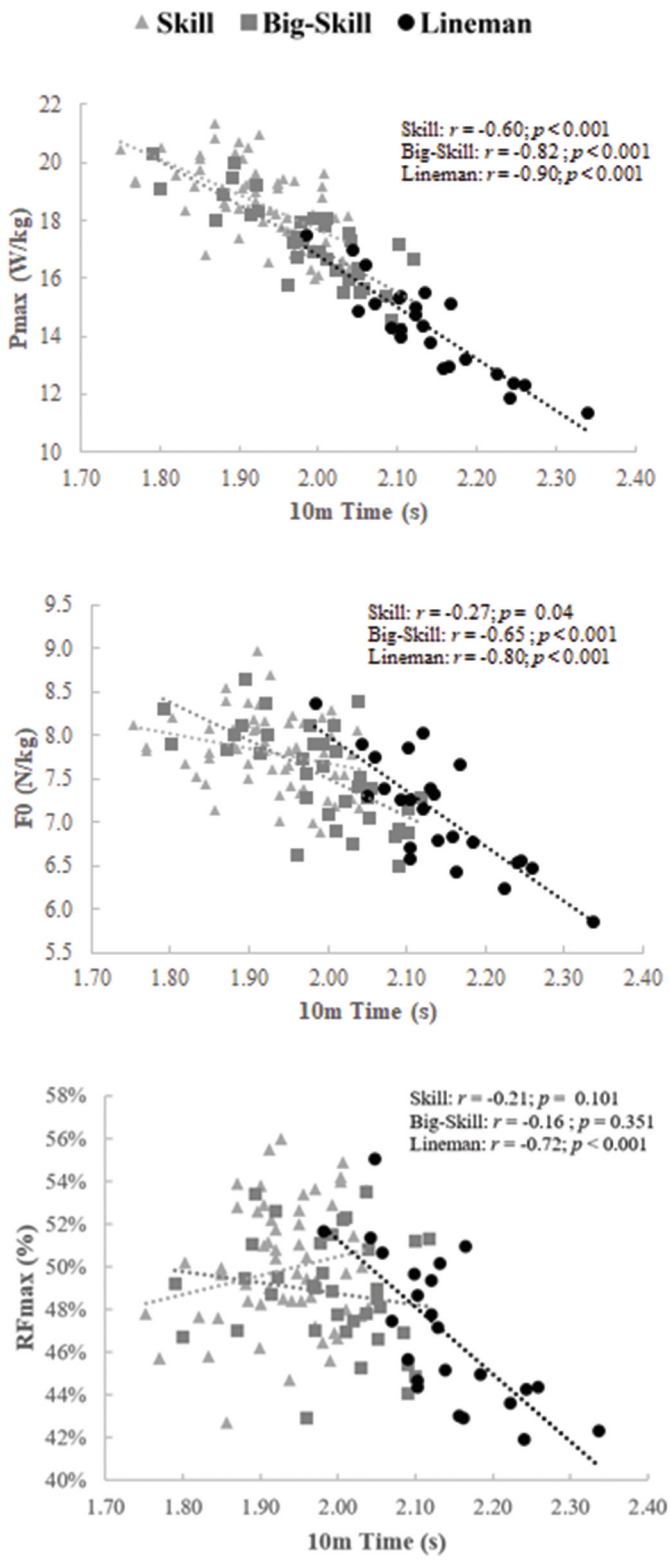

Figure 3. Relationships (Pearson's correlation coefficients) of 0–10-m split times and FVP variables Pmax, F0, and RFmax for Skill, Big-Skill, and Lineman groups. Gray triangles = Skill group, dark gray squares = Big-Skill group, and black circles = Lineman group.

Table 4. Relationships (Pearson's correlation coefficients and their 95% confidence intervals) of Pmax (W/kg) with spatiotemporal variables, joint kinematics at toe-off, and joint kinematics at touchdown.

	Variables	*r*	*p*-Value	Lower CL	Upper CL
Spatiotemporal values	**Contact Time (s)**	**−0.69**	**< 0.001**	**−0.77**	**−0.58**
	Takeoff velocity (m/s)	**0.65**	**< 0.001**	**0.54**	**0.74**
	Step velocity (m/s)	**0.58**	**< 0.001**	**0.45**	**0.69**
	Flight length (m/body length)	**0.56**	**< 0.001**	**0.42**	**0.67**
	Step frequency (Hz)	0.46	< 0.001	0.31	0.59
	Flight time (s)	0.42	< 0.001	0.26	0.55
	Step length (m/body length)	0.37	< 0.001	0.20	0.51
Joint position at toe-off	**Swing leg knee angle (°)**	**−0.67**	**< 0.001**	**−0.76**	**−0.56**
	Swing leg hip angle (°)	**−0.59**	**< 0.001**	**−0.70**	**−0.46**
	CM-toe distance (m/body length)	**−0.55**	**< 0.001**	**−0.66**	**−0.41**
	Stance leg thigh angle (°)	**−0.54**	**< 0.001**	**−0.65**	**−0.40**
	CM angle (°)	−0.47	< 0.001	−0.60	−0.32
	Swing leg thigh angle (°)	0.44	< 0.001	0.29	0.57
	Trunk angle (°)	−0.36	< 0.001	−0.51	−0.20
	Swing leg shank angle (°)	−0.34	< 0.001	−0.49	−0.17
Joint position at touchdown	**Stance leg shank angle (°)**	**−0.57**	**< 0.001**	**−0.68**	**−0.44**
	Swing leg shank angle (°)	**−0.57**	**< 0.001**	**−0.68**	**−0.44**
	Stance leg knee angle (°)	**−0.51**	**< 0.001**	**−0.63**	**−0.37**
	Stance leg foot angle (°)	−0.37	< 0.001	−0.51	−0.20
	Trunk angle (°)	−0.36	< 0.001	−0.50	−0.19
	Stance leg hip angle (°)	−0.33	< 0.001	−0.48	−0.17
	Touchdown distance (m/body length)	0.31	< 0.001	0.14	0.47
	CM angle (°)	−0.30	< 0.001	−0.46	−0.13

Bold indicates large (> 0.5) and significant (p < 0.001) relationship.

Table 5. Relationships (Pearson's correlation coefficients and their 95% confidence intervals) of F0 (N/kg) with spatiotemporal variables, joint kinematics at toe-off, and joint kinematics at touchdown.

	Variables	*r*	*p*-Value	Lower CL	Upper CL
Spatiotemporal values	**Step velocity (m/s)**	**0.59**	**< 0.001**	**0.46**	**0.69**
	Takeoff velocity (m/s)	**0.53**	**< 0.001**	**0.39**	**0.65**
	Contact time (s)	−0.47	< 0.001	−0.60	−0.32
	Step length (m/body length)	0.42	< 0.001	0.26	0.56
	Flight length (m/body length)	0.37	< 0.001	0.21	0.51
	Step frequency (Hz)	0.33	< 0.001	0.16	0.48
Joint position at toe-off	**CM-toe distance (m/body length)**	**−0.61**	**< 0.001**	**−0.71**	**−0.48**
	Stance leg thigh angle (°)	**−0.56**	**< 0.001**	**−0.67**	**−0.42**
	Swing leg hip angle (°)	**−0.52**	**< 0.001**	**-0.64**	**−0.37**
	CM angle (°)	**−0.51**	**< 0.001**	**−0.63**	**−0.36**
	Swing leg knee angle (°)	−0.48	< 0.001	−0.61	−0.33
	Trunk angle (°)	−0.38	< 0.001	−0.52	−0.21
	Swing leg thigh angle (°)	0.31	< 0.001	0.14	0.47
Joint position at touchdown	**Swing leg shank angle (°)**	**−0.51**	**< 0.001**	**−0.63**	**−0.37**
	Stance leg shank angle (°)	−0.37	< 0.001	−0.51	−0.21
	Trunk angle (°)	−0.37	< 0.001	−0.51	−0.20
	Knee angle (°)	−0.37	< 0.001	−0.51	−0.20
	Hip angle (°)	−0.35	< 0.001	−0.49	−0.18

Bold indicates large (> 0.5) and significant (p < 0.001) relationship.

of performance over short sprint distances (e.g., < 10-m) and vice versa for velocity-oriented profiles and longer sprint distances (e.g., > 20-m). Thus, the overall objective for sprint acceleration training is primarily to increase the value of Pmax and (to a lesser extent) align the FVP profile closer to the optimal orientation for the targeted sprint distance.

Additionally, F0 had a large to very large negative correlation with 0–10-m sprint times for the Lineman and Big-Skill groups, but not for the Skill group. The importance of body mass in relation to acceleration performance was illustrated in our results, as there was only a 3.65% difference for the values of F0 between the Big-Skill and Skill groups, but a 20.6% difference in body mass. It may be that for players under a certain threshold of body mass (e.g., < 106 kg, the mean body mass for the Big-Skill group in this study), a minimum level of F0 is necessary, after which achieving more horizontal force at high to very high velocities (e.g., V0) become increasingly associated with reduced sprint times at distances as low as 10-m. This would align with other findings where maximal velocity (Vmax) was strongly correlated with sprint times at both 9.1 and 18.3-m with a very large difference between groups with fast vs. slow Vmax, despite comparable acceleration patterns (Clark et al., 2019).

For RFmax, there was only a significant relationship with 0–10-m sprint times for the Lineman group and this

relationship was very large. This may coincide with the previous point of how players with large body masses may produce a higher ratio of vertical ground reaction impulses relative to the resultant force simply to support body weight and those capable of producing higher ratios of propulsive force with large body masses will portray elevated performances in sprint acceleration. Alternatively, the Skill and Big-Skill groups did not present a significant relationship between RFmax and 0–10-m sprint times, despite previous studies describing the importance of mechanical effectiveness on acceleration performance (Bezodis et al., 2019; Morin et al., 2011; Rabita et al., 2015) and an almost perfect correlation between RFmax and NAHEP ($r = 0.96$) (Bezodis et al., 2019). Measuring averaged RF values may have yielded different results than solely analysing RFmax, as Rabita et al. (2015) found an almost perfect correlation between averaged RF and 40-m sprint performance ($R^2 = 0.93$; $p < 0.001$). The importance of high net anteroposterior (horizontal) GRF for acceleration performance has previously been described, and there is a clear differentiation between elite sprinters and sub-elite sprinters in their ability to produce higher horizontal force at any given velocity (Morin et al., 2011, 2012; Rabita et al., 2015). Despite failing to find a direct significant relationship between measures of RFmax and the 0–10-m sprint segment in the Skill and Big-Skill groups, our data did reveal very large, significant relationships between RFmax and F0 ($r = 0.83$; $p < 0.001$) across the entire population, which is in line with previous studies (Jimenez-Reyes et al., 2021; Samozino et al., 2016) and provides insight into how force output is still largely explained by mechanical effectiveness.

These kinetic outputs may also play a role in prowess of sports-specific technical skills, as significant positive correlations were found between Pmax and efficiency of specific rugby skills in professional rugby players: namely duelling, tackling, and rucking (Glaise et al., 2023). The positive relationship with Pmax was primarily explained by higher levels of F0 and RFmax, indicating that horizontal force and mechanical effectiveness at low velocities are paramount to performance in these combative skills. These findings lend further support to the importance of F0 for the Lineman group in American football, as various forms of opponent engagement and disengagement through similar combative skills occur more frequently for these players in comparison to positions in the Skill and Big-Skill groups (Bayliff et al., 2019). Thus, there may be beneficial connections between non-specific linear sprint skills and game skills like those mentioned prior.

From a spatiotemporal perspective, Pmax was significantly associated with extending the stance leg further posterior to the CM (CM-toe distance) and achieving greater projection of the CM anteriorly through the air (flight length) after toe-off. The Lineman group spent more time on the ground and less time in the air, thereby exhibiting significantly reduced takeoff velocity and flight length compared to the Skill and Big-Skill groups. However, for measures of normalised step length and contact length, there was similarity among all groups. It seemed that the Lineman group was able to project the CM with similar relative displacement to the Skill and Big-Skill groups, but the step velocity was significantly slower. For step frequency, the Lineman group was significantly different only from the Skill group. However, had we looked at the change in step frequency from step to step rather than step-averaged values, the magnitude of differences between position groups may have been larger, as previous research indicates that initial acceleration performance is dependent upon large increases in step frequency (Nagahara et al., 2014). Additionally, the magnitude of step frequency has been described as an important determinant of step velocity (von Lieres et al., 2020).

At toe-off, Pmax had significant relationships with reduced hip and knee angles in the swing leg, as well as a reduced stance leg thigh angle relative to the horizontal axis. Compared to the Lineman group, the Skill and Big-Skill groups had significantly reduced swing leg hip and knee angles, indicating greater hip and knee flexion. Stance leg thigh angle and swing leg hip angle were also significantly associated with F0, as was CM angle at toe-off. In the latter case, projecting the CM at a reduced angle where the CM is closer to the ground during extension of the stance leg appears important for maximising F0. At touchdown, Pmax was significantly associated with reduced stance leg shank and knee angles and the swing leg shank angle being closer to the horizontal axis. The swing leg shank angle at touchdown was also significantly associated with F0, perhaps denoting the importance of interlimb 'scissors' coordination during the swing phase for achieving subsequent horizontal force application during the stance phase (Kakehata et al., 2023). Compared to the Skill and Big-Skill groups, the Lineman group had more extended knees at touchdown, resulting in larger angles of stance and swing leg shank. These 'negative' shank angles could result in larger touchdown distances, possibly increasing braking forces and negating acceleration performance (Hunter et al., 2005). This pattern may also be a direct result of the constraints imposed by having a larger body mass where the players in the Lineman group adapted their body position(s) to generate

sufficient vertical ground reaction forces to support their body weight while simultaneously aiming to achieve horizontal propulsion.

While exploring the relationships between kinetic and kinematic variables is useful for identifying trends and key performance indicators, especially when observed at the inter-group level, any search for an over-arching 'optimal' technique may prove futile. As an example, Wild et al. (2022) divided professional rugby union backs into four homogenous clusters based on the relationships between the subjects' step length/step frequency and contact time/flight time ratios over the first four steps of a sprint acceleration. The authors found no significant differences in sprint performance between clusters when based on NAHEP, signifying that an optimal technique for initial sprint performance does not exist in this population. Likewise, other studies have discussed inter-individual differences in coordination strategies during acceleration (Donaldson et al., 2022, 2023). Inter-athlete technique degeneracy may be another factor influencing the relationships between kinetic and kinematic variables, and this should be investigated further in the American football population.

Limitations

Future research could explore the relationships between sprint acceleration kinetics (Pmax and F0) and kinematics at the inter-group level, as the present study reported these relationships only across the entire study population. Similar analyses may consider increasing the number of players in the Lineman group, as some of the results presented here may show different levels of significance with a larger subject pool. Additionally, the results presented are from observations of Division I collegiate American football players, so a similar study design performed on professional American football players in the NFL might yield different results with a more homogenous population of elite-level players. Another consideration for future research would be to analyse relationships at the intra-group level, perhaps comparing intra-positional groupings of the fastest and slowest performers.

Conclusion

The primary finding of this study is that Pmax had the largest relationship with initial sprint acceleration performance of 0–10-m for all position groups and may serve as an appropriate key performance indicator of initial acceleration. The relative importance of incorporating F0 as a key performance indicator for initial acceleration appears more important as player body mass increases (e.g., ≥106 kg). From a spatiotemporal perspective, reduced contact time and greater takeoff velocity, step velocity, and flight length had large significant correlations with Pmax. At toe-off, Pmax had large significant correlations with joint positions of greater flexion of the swing leg (e.g., reduced hip and knee angles) and extension of the stance leg (e.g., reduced thigh angle and toe further behind the CM). At touchdown, reduced angles of the stance leg knee and shank, along with a swing leg shank angle closer to the horizontal axis, all had large significant correlations with Pmax. This study is the first to describe the kinetic and kinematic variables and how the relationships between them may impact initial acceleration performance in American football. The information presented may provide more context for investigating effective monitoring and training interventions to enhance initial acceleration in the American football player population.

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